

Matt Wilson

Assistant Professor in Computer Science

CentraleSupélec: Université Paris-Saclay – Paris, France

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Assistant Professor in Computer Science specialising in applied category theory and quantum foundations. My research develops categorical foundations for higher-order operations, with applications to quantum information theory, quantum causality, and the broader study of the agent–environment interaction. I am actively investigating the mathematical structure of theories of higher-order operations, including how to organise them into higher-categories and frame equivalence between them. I am furthermore developing their applications to infinite-dimensional, field-theoretic, and generalised probabilistic systems, along with reward-based models of planning and learning in quantum and multi-agent systems.

Publication list: Google Scholar

Employment History

CentraleSupélec - Université Paris-Saclay

France

Assistant Professor

September 2024 – Present

Research in computer science, formal methods, and quantum information, alongside undergraduate and graduate teaching in computer science.

University College London

UK

EPSRC Doctoral Prize Fellowship

May 2023 – August 2024

Post-doc at the PPLV (Programming Principles, Logic, and Verification) group in the Department of Computer Science, working on the categorical semantics of higher-order quantum operations.

SnapRevise

UK

Physics Content Creator

July 2017 – January 2018

I designed and produced short-form video lessons on GCSE Physics for EdTech startup SnapRevise.

Education

University of Oxford

UK

PhD, Computer Science, Pass with minor corrections

Sep 2019 – Nov 2023

Thesis: “Categorical Frameworks for Supermaps and Causality”.

University College London

UK

Master's Degree, Delivering Quantum Technologies, Merit.

Sep 2018 – Sep 2019

University of Oxford

UK

MMathPhys, Mathematical and Theoretical Physics, Distinction.

Oct 2013 – Jun 2017

Selected Publications and Preprints

1: J. Hefford, M. Wilson. *A BV Category of Spacetime Interventions*. arXiv preprint arXiv:2502.19022, 2025

2: J. Hefford, M. Wilson. *A Profunctorial Semantics for Quantum Supermaps*. In Proceedings of

LICS 2024.

3: M. Wilson, G. Chiribella, A. Kissinger. *Quantum Supermaps are Characterized by Locality*. Quantum, 10, 2013, 2026

Publications

1: J. Hefford, M. Wilson. *Decoherence to quantum theory from a causally indefinite post-quantum theory*. Phys. Rev. A 113, 042433, 2026

2: M. Wilson, G. Chiribella, A. Kissinger. *Quantum Supermaps are Characterized by Locality*. Quantum, 10, 2013, 2026

3: J. Hefford, M. Wilson. *A Profunctorial Semantics for Quantum Supermaps*. In Proceedings, LICS 2024.

4: P. Arrighi, A. Durbec, M. Wilson. *Quantum networks theory*. Quantum, 8, 1508, 2024

5: M. Wilson, G. Chiribella. *Causality in higher order process theories*. In Proceedings QPL 2021, EPTCS 343, 2021, pp. 265-300.

6: O Higgott, M Wilson, J Hefford, J Dborin, F Hanif, S Burton, D E Browne. *Optimal local unitary encoding circuits for the surface code*. Quantum, 5, 517, 2021.

7: G. Chiribella, M. Wilson, HF. Chau. *Quantum and classical data transmission through completely depolarizing channels in a superposition of cyclic orders*. Physical Review Letters 127 (19), 190502, 2021.

8: M. Wilson, J. Hefford, G. Boisseau, V. Wang. *The Safari of Update Structures: Visiting the Lens and Quantum Enclosures*. In Proceedings ACT 2020, EPTCS 333, 2021, pp. 1-18.

9: M. Wilson, G. Chiribella. *A Diagrammatic Approach to Information Transmission in Generalised Switches*. In Proceedings QPL 2020, EPTCS 340, 2021, pp. 333-348.

10: H. Kristjánsson, G. Chiribella, S. Salek, D. Ebler, M. Wilson. *Resource theories of communication*. New Journal of Physics 22 (7), 073014, 2020.

Preprints

11: M. Wilson. *Categories of tagged lenses*. arXiv preprint arXiv:2607.25706, 2026

12: M. Wilson, J. Hefford. *Higher-Order Quantum Objects are Strong Profunctors*. arXiv preprint arXiv:2603.11221, 2026

13: M. Wilson. *Higher-order circuits*. arXiv preprint arXiv:2602.18701, 2026

14: M. Wilson. *Agent policies from higher-order causal functions*. arXiv preprint arXiv:2512.10937, 2025

15: G Agrawal, M. Wilson. *Deriving the Generalised Born Rule from First Principles*. arXiv preprint arXiv:2511.21355, 2025

16: O. Mestoudjian, M. Wilson, A. Vanrietvelde, P. Arrighi. *Picturing general quantum subsystems*. arXiv preprint arXiv:2511.09494, 2025

17: J. Hefford, M. Wilson. *A BV Category of Spacetime Interventions*. arXiv preprint arXiv:2502.19022, 2025

18: J H Selby, M E Stasinou, M Wilson, B Coecke. *Generalised Process Theories*. arXiv preprint arXiv:2502.10368, 2025

- 19: M. Wilson, N. Ormrod. *On the Origin of Linearity and Unitarity in Quantum Theory*. arXiv preprint arXiv:2305.20063, 2023
- 20: M. Wilson, G. Chiribella. *Free Polycategories for Unitary Supermaps of Arbitrary Dimension*. arXiv preprint arXiv:2207.09180, 2022
- 21: P. Arrighi, A. Durbec, M. Wilson. *Generalised tensors and traces*. arXiv preprint arXiv:2202.11340, 2022
- 22: M. Wilson, G. Chiribella. *A Mathematical Framework for Transformations of Physical Processes*. arXiv preprint arXiv:2204.04319, 2022
- 23: M. Wilson, A. Vanrietvelde. *Composable constraints*. arXiv preprint arXiv:2112.06818, 2021
- 24: J. Hefford, V. Wang, M. Wilson. *Categories of Semantic Concepts*. arXiv preprint arXiv:2004.10741, 2020

Funding and Awards

- Awarded an EPSRC Doctoral Prize Fellowship [Grant number EP/W524335/1] to develop my research proposal on the “Categorical Foundations of Quantum Transformations” (1+1 years of postdoctoral salary and conference travel).
- Collaborative grant participation through the Without Spacetime consortium.
- Offered the Royal Society JSPS Postdoctoral Fellowship (declined on taking up current post)
- Start-up support upon taking on a permanent position at CentraleSupélec / University of Paris-Saclay

Course Preparation Experience

- Quantum computation: I have prepared a short 9-hour course on quantum computation for students on the AIDAMS (AI and Data Management Science) degree at CentraleSupélec. This course will be delivered in academic year 2026/2027. In academic year 2027/2028 this is scheduled to become a full 18-hour course.
- SnapRevise: At educational startup SnapRevise I designed, prepared, and delivered a bite-size course on GCSE physics.

Lecturing Experience

- Artificial Intelligence: ($\approx$ 30 students - 15 hours) (April 2025 – May 2025)
- Artificial Intelligence: ($\approx$ 30 students - 12 hours) (April 2026 – May 2026)
- Information Systems and Programming: ($\approx$ 200 students - 6 hours) (September - October 2025)

Tutorial Class Experience

The following amounts to approximately 300 hours of teaching service at the undergraduate and graduate level, the majority of which have been provided within the last 1.5 years.

- Quantum Processes and Computation: Tutorials ($\approx$ 10 students - 6 hours) (2021)
- Quantum Information: Tutorials ($\approx$ 10 students - 6 hours) (2022)
- Advanced Algorithms: ($\approx$ 20 students - 6 hours) (September - October 2024)
- Information Systems and Programming: Tutorials ($\approx$ 30 students - 15 hours) (September - October 2024)

- Coding Weeks: Lab Supervision ($\approx$ 120 Students between 4 Supervisors, 45 hours) (November 2024)
- Algorithms and Complexity ($\approx$ 30 students - 20 hours) (November 2024 – January 2025)
- Programming for Data Science: Tutorials ($\approx$ 30 students - 24 hours) (January 2025 – February 2025)
- Introduction to Algorithms: Tutorials ($\approx$ 30 students - 10 hours) (March 2025 – April 2025)
- Quantum Machine Learning: Project Supervision ($\approx$ 10 students - 5 hours) (March 2025 – April 2025)
- Introduction to Artificial Intelligence: Tutorials ($\approx$ 10 students - 10 hours) (April 2025)
- Fundamentals of Programming: Tutorials ($\approx$ 30 students - 10 hours) (May 2025)
- Artificial Intelligence: Tutorials ($\approx$ 30 students - 15 hours) (April 2025 – May 2025)
- Introduction to Computer Science: Tutorials ($\approx$ 30 students - 20 hours) (September 2025 - December 2025)
- Information Systems and Programming: Tutorials ($\approx$ 30 students - 12 hours) (September 2025 - December 2025)
- Advanced Algorithms: Tutorials ($\approx$ 20 students - 6 hours) (September 2025 - October 2025)
- Quantum Compilation: Tutorials ($\approx$ 10 students - 6 hours) (September 2025 - October 2025)
- Coding Weeks: Lab Supervision ($\approx$ 70 Students between 2 Supervisors, 48 hours) (October/November 2025)
- Coding Weeks: Lab Supervision ($\approx$ 120 Students between 4 Supervisors, 48 hours) (November 2025)

Research Supervision and Mentorship

- Co-supervision of PhD student Octave Mestoudjian at Université Paris-Saclay (October 2024 – present)
- Supervision of research engineer Gaurang Agrawal at Université Paris-Saclay (December 2025 – May 2026)
- Supervision of the internship of Mohammed Boudjemaa at Université Paris-Saclay (June 2026 – Present)
- Co-supervision of the internship of François-Elie Ingwer at Université Paris-Saclay (May 2025 – August 2025)
- Co-supervision of the master's project of Kathleen Barsse at University of Oxford (Jan 2022 – July 2022)

Invited Talks

Higher-Order Quantum Structures (HOAQS) Workshop:

- "Theory and Applications of Categorical Supermaps" (2025)

Quantum TUT (Toyohashi University of Technology) Workshop:

- "Categorical Quantum Supermaps" (2024)

Plenary Talks

Quantum Physics and Logic (QPL):

- "Quantum Supermaps are Characterized by Locality" (2023)

Contributed Talks

Logic in Quantum Computer Science:

- "Higher-order Circuits" (2026)
- "Higher-order quantum objects are strong profunctors" (2026)

Quantum Physics and Logic (QPL):

- "A Diagrammatic Approach to Information Transmission in Generalized Switches" (2020)
- "Causality in Higher Order Process Theories" (2021)
- "A Mathematical Framework for Transformations of Physical Processes" (2022)
- "Polycategories of Supermaps on Monoidal Categories" (2023)
- "On the Origin of Linearity and Unitarity in Quantum Theory" (2024)

Applied Category Theory (ACT):

- "The Safari of Update Structures, Visiting the Quantum and Lens Enclosures" (2020)
- "Composable Constraints" (2021)
- "Quantum Supermaps are Characterized by Locality" (2022)
- "Polycategories of Supermaps on Monoidal Categories" (2023)
- "Supermaps on generalized theories" (2026)
- "Agent policies from higher-order causal functions" (2026)

Observers and causality in quantum gravity:

- "The spatio-temporal logic of agents, actions, and observations" (2026)

Causalworlds:

- "Supermaps on generalised theories" (2026)
- "Agent policies from higher-order causal functions" (Flash talk, 2026)

Time in quantum theory:

- "Higher-order processes beyond finite dimensional quantum theory" (2026)

Symposium on Compositional Structures (SYCO):

- "Composable Constraints" (2021)

Taming Quantum Causality:

- "Categorical Quantum Supermaps" Jan (2025).
- "Supermaps and process matrices on generalised theories" Jun (2025).
- "Agent Policies from Higher-Order Causal Processes" Jan (2026).

Reviewing and Refereeing

- Journal peer-review: *Quantum*, *Journal of Mathematical Physics*, *Foundations of Physics*
- Conference peer-review: *Quantum Physics and Logic* (QPL)
- Workshop submission review: *Causalworlds*

Organisational Service

- Programme Committee member, *Causalworlds* (2024, 2026)
- Organising committee member of *Quantum Physics and Logic* (QPL) 2020.

Departmental Service

- Conducting university entrance interviews

- Study-abroad advising and sign-off
- Student references

Publicity and outreach

- Our paper "*Decoherence to quantum theory from a causally indefinite post-quantum theory*" was the topic of a New Scientist article titled "*QBox theory may offer glimpse of reality deeper than quantum realm*"
- For one year I ran the QISS (Quantum Information Structure of Spacetime) YouTube seminar series

Diversity and inclusion

- Participation in a 6 hour course "Captive" on teaching in an international classroom
- Participation in a mock university interview day with Teach First.

Private tutoring experience

- Real analysis, Number theory, and Group theory. Approximately 20 hours total.